Practical No : 16

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Q.16) Write a C program for fractional knapsack problem.

```
#include <stdio.h>
struct Item {
    int weight;
    int profit;
    float ratio;
};
int main() {
    int n, capacity;
    int i, j;
    float totalProfit = 0;

    printf("Enter number of items: ");
    scanf("%d", &n);

    struct Item items[n];

    printf("Enter weight and profit of each item:\n");
    for (i = 0; i < n; i++) {
        printf("Item %d: ", i + 1);
        scanf("%d %d", &items[i].weight, &items[i].profit);

        items[i].ratio = (float)items[i].profit / items[i].weight;
    }

    printf("Enter knapsack capacity: ");
    scanf("%d", &capacity);
```

```
// Sort items by profit/weight ratio in descending order
for (i = 0; i < n - 1; i++) {
    for (j = i + 1; j < n; j++) {
        if (items[i].ratio < items[j].ratio) {
            struct Item temp = items[i];
            items[i] = items[j];
            items[j] = temp;
        }
    }
}

// Select items
for (i = 0; i < n; i++) {
    if (capacity >= items[i].weight) {
        // Take the entire item
        capacity -= items[i].weight;
        totalProfit += items[i].profit;

        printf("Item %d: 100%% taken\n", i + 1);
    }
    else {
        // Take the fraction that fits
        float fraction = (float)capacity / items[i].weight;
        totalProfit += items[i].profit * fraction;

        printf("Item %d: %.2f%% taken\n", i + 1, fraction * 100);

        capacity = 0;
        break;
    }
}
```

```
    printf("\nMaximum Profit = %.2f\n", totalProfit);


    return 0;
}
```

Output:

